Functional Requirements

1. The program will read the language BasicML and interpret it to execute the corresponding operations
2. The program will allow the user to upload their instructions to the file if the file is in a .txt format and the instructions are in number format with a + or - in front of them.
3. The program will halt when the instruction is 4300
4. The program will save the instructions in memory starting from address 00
5. The maximum memory will be 99
6. The program will iterate through each instruction in memory.
7. The program can manipulate the accumulator if the instructions require it to be compared or changed.
8. The program will display an error message if the user input is not an integer
9. The program will be able to load and store integers into different sections of the memory or the accumulator.
10. The program will be able to do basic mathematical operations on the integers given by the user or stored in the data.
11. The program will be able to branch to other locations of memory when given the instruction 40.
12. The program will be able to branch to other locations of memory depending on the value of the accumulator if the instructions are 41 or 42.
13. The program will be able to write and read inputs from the user/memory.
14. The program will wipe its memory once a new set of instructions is executed.
15. The program will wipe the accumulator value once a new set of instructions is executed.

Non-Functional Requirments

1. The program will be able to handle instructions that can be up to 100 instructions in length
2. The program will be able to run the instructions and execute them within 5 seconds.
3. The program will prompt the user to load a specific instruction file before the program fully executes.